

Synoptic CB: Lateral Transport Campaigns 2024

GCW Discrete DIC Samples

2025-09-17

Contents

0.1	Import Data Functions	2
0.2	Import Sample Data	2
0.3	Assessing Standard Curves	3
0.4	CRM Check - Don't run chunk if no CRMs run	4
0.5	Assess Check Standards	5
0.6	Assess Blanks	5
0.7	Sample Flagging - Are samples Within the range of the curve?	6
0.8	Visualize Data by Plot	7
0.9	Convert data from mg/L to uMoles/L	8
0.10	Export Processed Data	8

```
##Setup - Change things here & write any notes
```

```
#identify section  
cat("Setup Information")
```

```
## Setup Information
```

```
##### Run information - PLEASE CHANGE  
Date_Run = "07/26/24" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "No duplicates on this run." #any notes from the run  
samples <- c("LatI", "GCW") #whatever identifies your samples within the same names  
samples_pattern <- paste(samples, collapse = "|")  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_" #what did you name your check standards?  
crm_name = "CRM|crm" #what did you name your CRMS?  
  
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = c("Raw_GCW_Data/GCW_DIC.txt" )  
  
#file path and name for raw all peaks file  
raw_allpeaks_name = "Raw_GCW_Data/GCW_DIC_allpeaks.txt"  
  
#file path and name of processed data file  
processed_file_name = "Processed_GCW_Data/GCW_DIC_processed.csv"  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
  #Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv" - no metadata file for these  
  
#qaqc log file path for this year  
Log_path = "Raw_GCW_Data/COMPASS_Synoptic_DIC_QAQClog_2025.csv" #no log file for these
```

```
##Set Up Code
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:  
## * ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,  
## e.g.:  
##   dat <- vroom(...)  
##   problems(dat)
```

```
## # A tibble: 6 x 3
##   sample_name          ic_raw run_datetime
##   <chr>              <dbl> <chr>
## 1 LatT_GCW_UP1_FW1_20240715_1500    10.6 7/26/2024 7:57:15 PM
## 2 LatT_GCW_UP2_FW1_20240715_1515    14.0 7/26/2024 8:10:13 PM
## 3 LatT_GCW_WC1_20240715_1600     117. 7/26/2024 8:27:04 PM
## 4 LatT_GCW_WC2_20240715_1630     124. 7/26/2024 8:44:52 PM
## 5 LatT_GCW_WE1_20240715_1650     82.5 7/26/2024 9:07:15 PM
## 6 LatT_GCW_WE2_20240715_1730     83.7 7/26/2024 9:29:39 PM
```

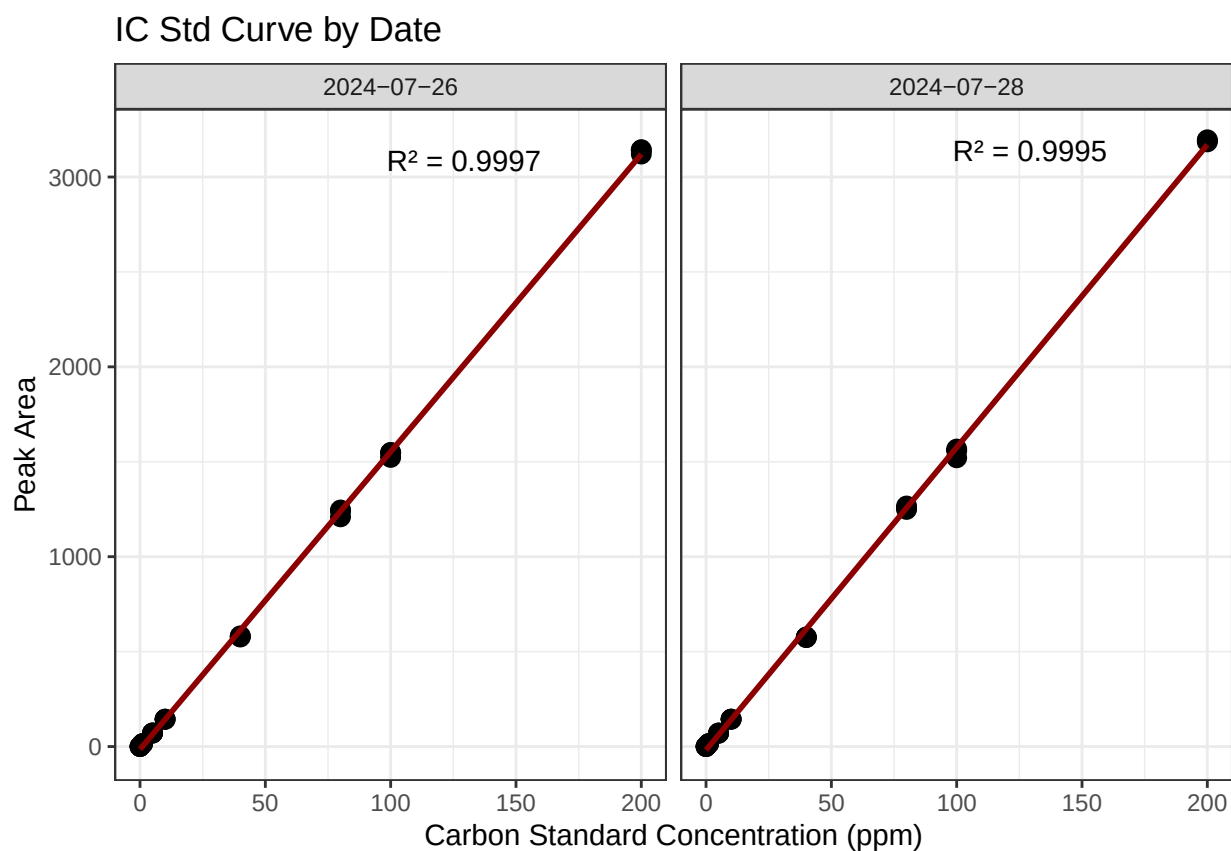
0.3 Assessing Standard Curves

```
## Assess the Standard Curves
```

```
## New names:
## * ' ' -> '...18'
```

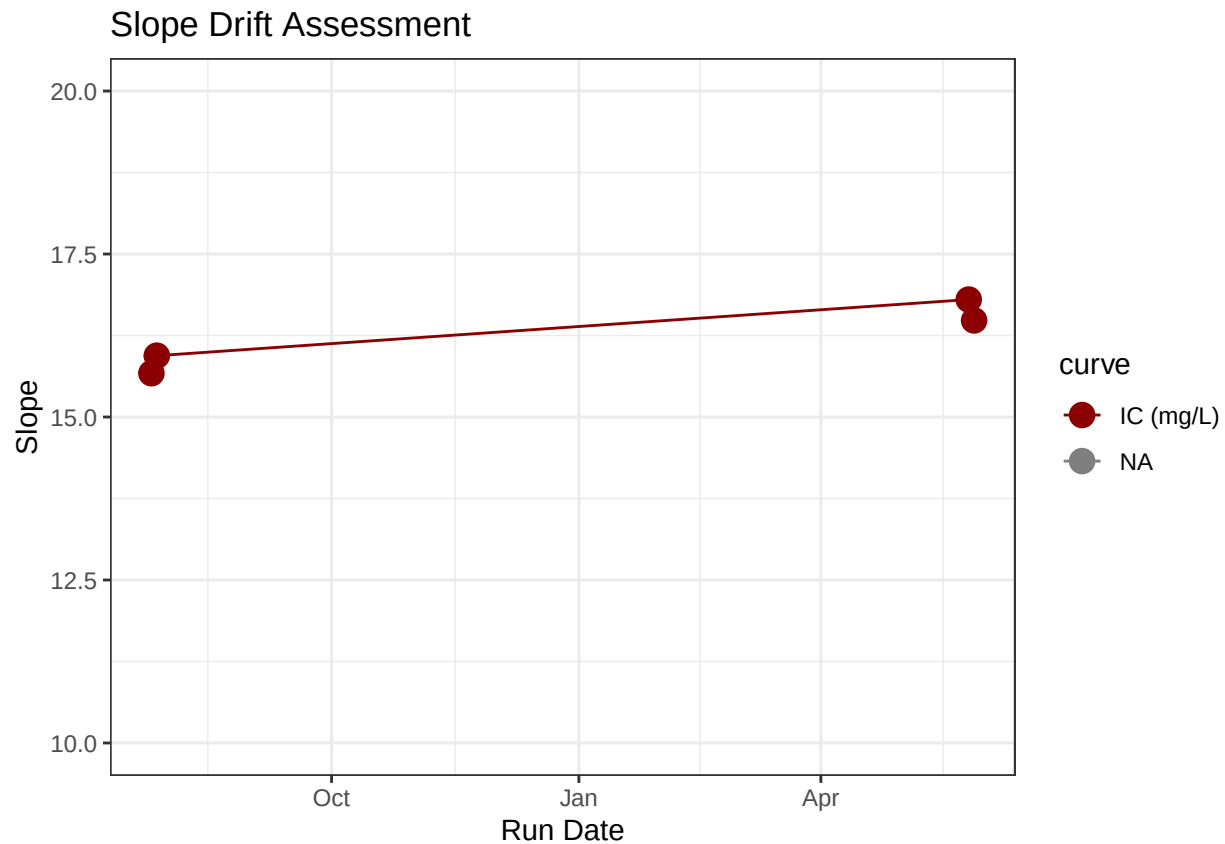
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
## [1] "IC Curve r2 GOOD"
```

0.4 CRM Check - Don't run chunk if no CRMs run

```
## Assess the CRMs
```

```
## New names:
## * '' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## [1] "IC crm has a % Difference <25% of expected - PROCEED"
```

```
## Run mean = 20.71143
```

```
## Expected = 22.19
```

0.5 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
```

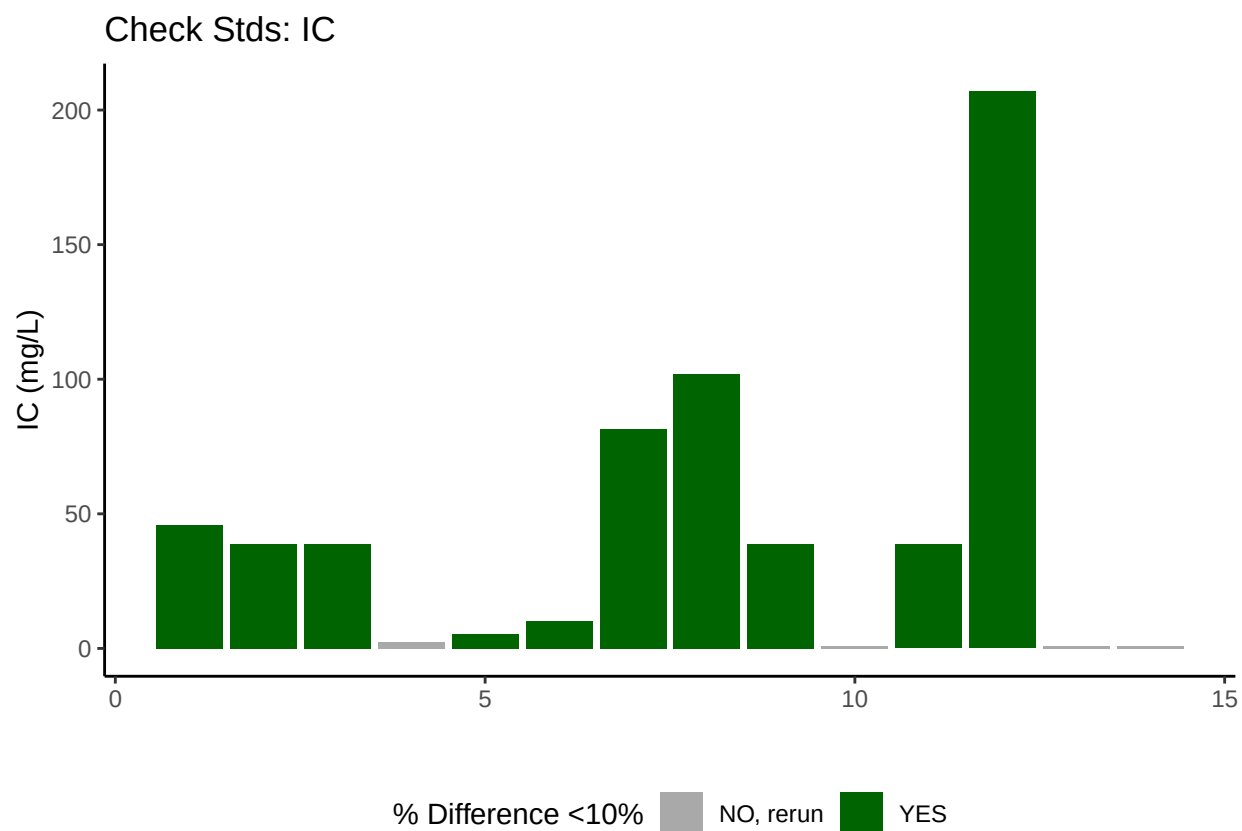
```
## * `` -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
```

```
## e.g.:
```

```
##   dat <- vroom(...)
```

```
##   problems(dat)
```



```
## [1] ">60% of IC Check Standards are within range of expected concentration"
```

0.6 Assess Blanks

```
## Assess Blanks
```

```
## New names:
```

```
## * `` -> '...14'
```

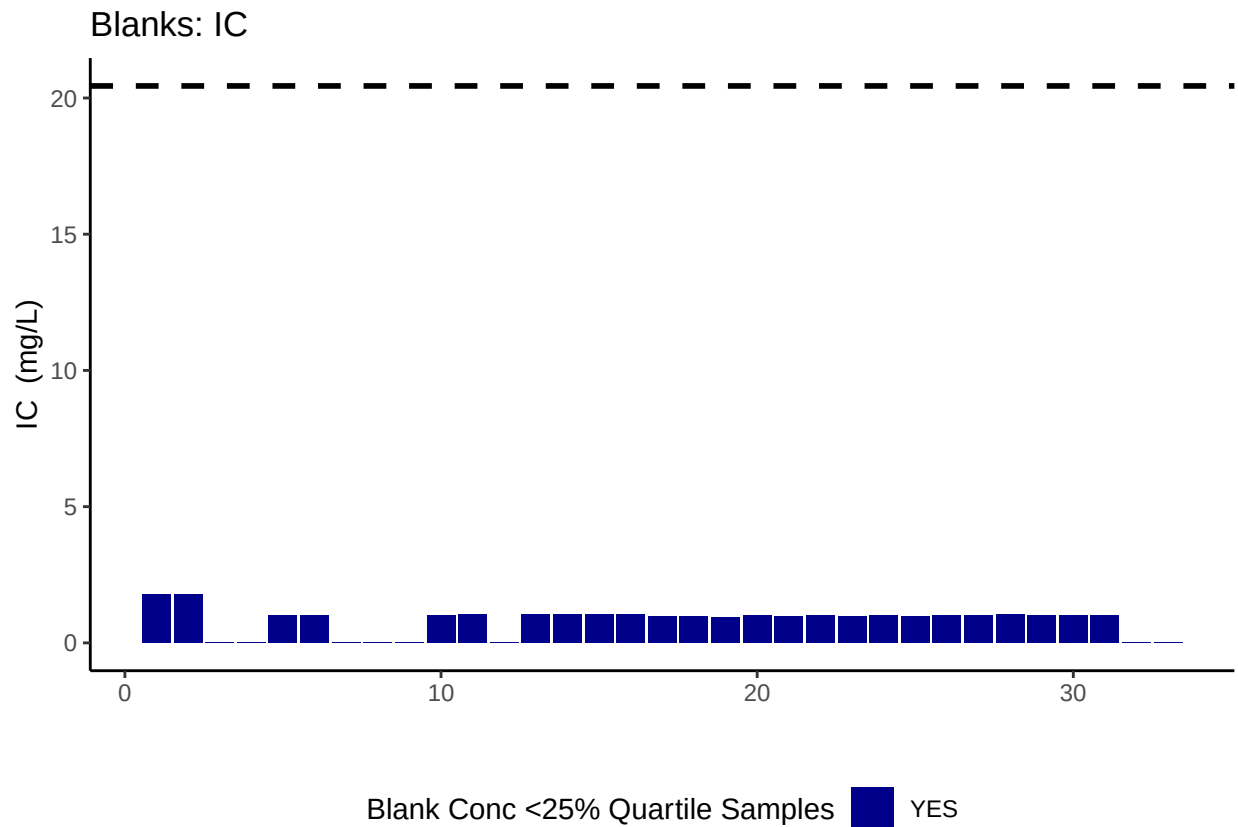
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
```

```
## e.g.:
```

```
##   dat <- vroom(...)
```

```
##   problems(dat)
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.8096515
```

0.7 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```

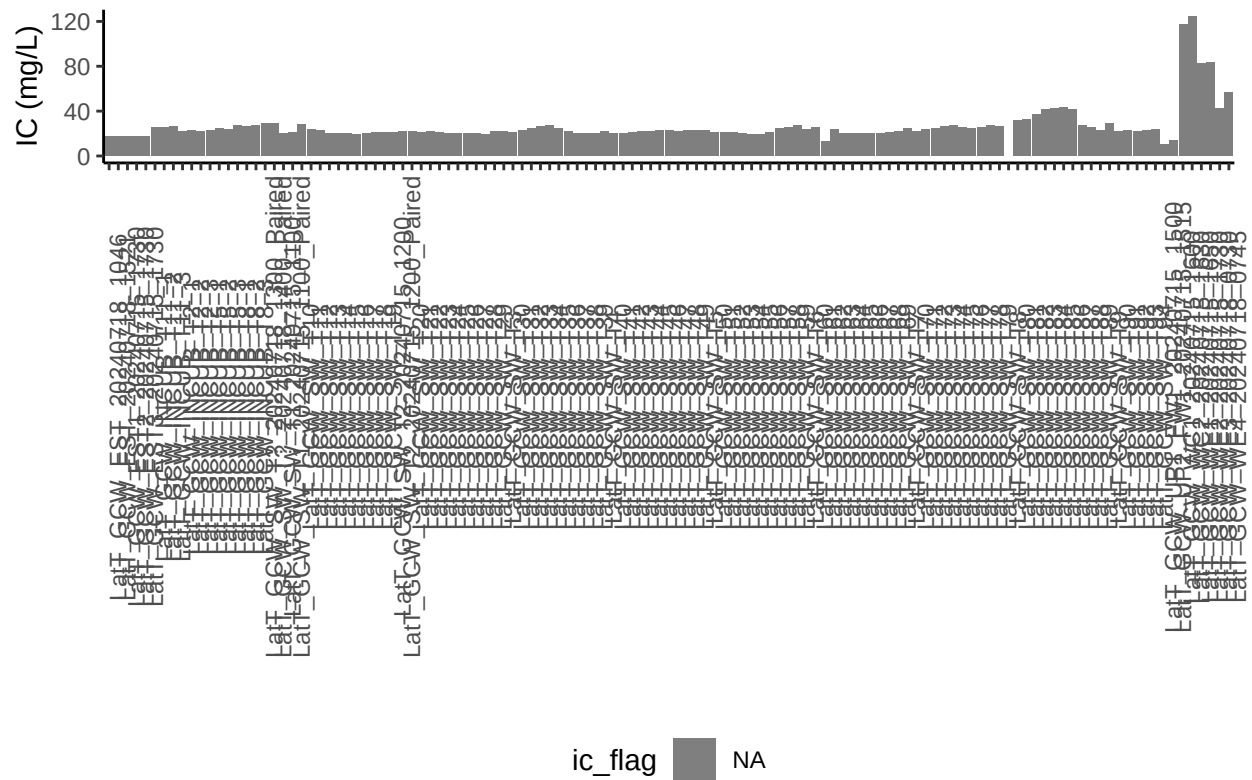
```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
## width unknown for character 0xd
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
## width unknown for character 0xd
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## font width unknown for character 0xd
```

```
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## font width unknown for character 0xd
```

C: Grey = Within Range of Curve



0.8 Visualize Data by Plot

Visualize Data

```
## Warning in rbind(c("LatT", "GCW", "UP1-FW1", "20240715", "1500"), c("LatT", :
## number of columns of result is not a multiple of vector length (arg 1)
```



```
## 5      7/26/2024 9:07:15 PM No duplicates on this run.  
## 6      7/26/2024 9:29:39 PM No duplicates on this run.
```

```
#end
```